5G Handover Optimization using Machine Learning

A Predictive Approach based on the DoNext Dataset

Integrated Project (PI) Report

Academic Year: 2025-2026

Realized by:

Zakaria Zarrouk
Mohamed Aziz
Hamza Rezgui
Hajer Maatoug
Ghaya Korbi
Wiem Ben Hassine

Class: 4ème Data

April 2, 2026

Contents

Scientific Context **3**
The Handover Challenge . . . 3
Project Scope . . . 4
Methodology . . . 4

Phase 1: Business Understanding **5**
Problem Formulation (Business View) . . . 5
Study and Critique of Existing Solutions . . . 5
Proposed Direction (Business-Level Summary) . . . 6
Business Objectives (BO) and Business KPIs . . . 6
Data Science Objectives (DSO) . . . 8
BO–DSO Mapping . . . 9
Justification of the BO–DSO links . . . 9
Sustainable Development Goals (SDG) Alignment . . . 10

Phase 2: Data Understanding & Exploration **11**
Dataset Overview . . . 11
Scenario Differences (What changes across the 3 datasets) . . . 12
Targets (QoS outcomes) and justification . . . 13
Feature dictionary . . . 13
Feature selection protocol (evidence-based) . . . 14
Dataset Component Description . . . 14
Comprehensive Correlation Analysis . . . 15
Key feature selection (what we keep for analysis and why) . . . 16
Data Availability and Quality Assessment . . . 17
Labels and Target Definitions . . . 18
Feature Groups . . . 18
Traceability: Feature Groups vs. DSOs . . . 19
Key Variables Description . . . 19
Descriptive Statistics . . . 19
Exploratory Data Analysis (EDA) . . . 20
Signal Strength Distribution (RSRP) . . . 20
Latency Distribution (Mean RTT) . . . 20
Velocity Distribution . . . 22
Serving vs. Neighboring Cell Analysis . . . 22
Data Quality Assessment . . . 24
Key Findings Summary . . . 24

Phase 3: Data Preparation **25**
3.1 Initial Feature Pool 25
3.2 Quality Filtering 26
3.3 Redundancy Pruning 27
3.4 Outlier Treatment and Visualization 28
3.5 Feature Selection and Engineering 28
3.6 Standardization 28
3.7 Train/Test Split 29

Phase 4: Modeling **30**
Model Selection 30
Training Strategy 31

Phase 5: Evaluation **34**
Performance Metrics 34
Results Analysis 35

Phase 6: Deployment **37**
Deployment Architecture 37
Continuous Learning 37

General Conclusion **50**

Scientific Context

The telecommunications landscape is undergoing a paradigm shift with the massive deployment of 5G networks. Unlike its predecessors, 5G is not merely about higher peak speeds; it is designed to support new service categories with strict performance requirements:

- **Mission-critical low-latency and high-reliability services** enabling applications where communication stability is essential. For example, recent discussions and trials around *5G-enabled telesurgery* show how low-latency and high-reliability connectivity can support remote robotic procedures, where the network must provide stable real-time responsiveness to avoid safety risks [3, 4].
- **Massive Internet of Things (mMTC / massive connectivity)** supporting very large numbers of connected devices in smart-city and industrial environments. Practical smart-city visions rely on dense deployments of sensors (traffic monitoring, environmental sensing, smart lighting, etc.), which motivates 5G designs targeting massive device connectivity and efficient signaling at scale [2].
- **Stringent Quality of Service (QoS)** requirements for consumer and interactive services. For instance, Netflix recommends a minimum connection speed of **15 Mbps** to stream Ultra HD (4K), illustrating the bandwidth stability expected by end users for high-quality video consumption [5].

However, maintaining these demanding standards in dense urban environments poses a significant challenge, particularly for **mobility management**—the process of keeping users connected as they move between cell towers. Under mobility, handovers can introduce short service interruptions, latency spikes, or unstable cell switching, which directly impacts user experience and SLA compliance. This report focuses on data-driven handover optimization to improve service continuity, QoS stability, and operational efficiency.

The Handover Challenge

One of the most critical processes in cellular networks is the **Handover (HO)**—the mechanism by which a user's session is transferred from one cell tower to another. Traditional handover algorithms, which rely on static signal thresholds, are increasingly becoming bottlenecks. They are reactive by nature, often leading to:

- **Call drops:** A passenger on a high-speed train (300 km/h) experiences 3–5 call interruptions per 50 km journey.[7].
- **Latency spikes:** Online gamers experience lag spikes of 200–500 ms during cell transitions.[6].

- **Ping-Pong effect:** A user at the edge of two cells switches between them 10+ times per minute unnecessarily.[1].

Project Scope

This project, realized within the framework of the **Integrated Project (PI)** for the academic year **2025–2026**, addresses these challenges by proposing an AI-driven approach to Self-Organizing Networks (SON). Our team aims to replace legacy rules with intelligent, predictive models capable of anticipating network conditions.

Methodology

In this report, we follow the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) methodology to structure our work. We begin by defining the business objectives, proceed to analyze the "DoNext" dataset, and finally design and evaluate Machine Learning models optimized for mobility management in 4G/5G networks. This structured approach ensures that our technical solutions are directly aligned with the operational goals of modern network providers.

Phase 1: Business Understanding

Context (Operator & Customer View)

In a competitive 5G market, users expect **continuous connectivity while moving** (train, car, metro, walking). For network operators, mobility performance directly influences **customer satisfaction** and **churn**. Mobility behavior also impacts **operational efficiency** through signaling load and repeated handovers. This project addresses mobility management in **business terms**: reduce customer-perceived interruptions, improve low-latency service stability under mobility, and reduce unnecessary overhead.

Problem Formulation (Business View)

When a customer moves, the network transfers the connection from one cell to another through a **handover (HO)**. In dense and dynamic environments (buildings, tunnels, shadowing, congestion), late or unstable HO decisions often result in:

- **Interruptions:** the customer experiences **call drops**, frozen video/voice, or short connection breaks.
- **Instability (ping-pong):** repeated back-and-forth switching between two cells, creating unnecessary network activity.
- **QoS degradation:** throughput drops and latency spikes, risking enterprise SLA violations.

Study and Critique of Existing Solutions

Traditional mobility management is typically rule-based (fixed thresholds and timers). While robust in stable conditions, it has business limitations in modern 5G:

1. **One-size-fits-all tuning:** static parameters do not adapt to different speeds, loads, or environments.
2. **Decisions taken too late:** reacting after quality has degraded increases interruption risk.
3. **Experience blind spots:** selecting a target cell based only on signal strength does not ensure stable post-handover experience.
4. **Limited learning loop:** recurring problem zones are not automatically identified and mitigated.

Proposed Direction (Business-Level Summary)

We introduce an **anticipate–recommend–validate** logic using operational measurements:

- **Anticipate risk early:** raise an early warning before quality breaks.
- **Recommend a stable target:** choose the next cell based on expected experience stability (not only instantaneous radio).
- **Validate before committing:** estimate post-HO QoS to avoid SLA-breaking switches.
- **Learn recurring issues:** identify high-risk contexts/zones to guide targeted optimization.

Business Objectives (BO) and Business KPIs

The business objectives below are written in a **marketing-friendly** way (what we promise + why it matters), while still being measurable through operator KPIs.

1. **BO1: Deliver a Seamless Mobile Experience (Zero-Frustration Connectivity)**
   - *What we promise:* smooth, uninterrupted connectivity for users while they move.
   - *Why the customer cares:* customers do not understand "handover failures"; they understand **calls dropping**, **internet freezing**, and **lag in video/gaming**. Every interruption increases frustration and churn risk.
   - *Business impact:* fewer complaints, higher satisfaction, stronger brand reputation, reduced churn.
   - *Clear example:* A passenger in public transport joins a video meeting; repeated HO issues cause freezes and occasional call drops. The objective is a **continuous perceived experience**.
   - *KPIs (explained):*
     - **Call/Session Drop Rate:** percentage of sessions that terminate unexpectedly (direct customer symptom).
     - **Handover Failure Rate:** percentage of HO attempts that fail to complete (key network root cause of drops/interruptions).
   - *Simple value message: "We make the network feel invisible — users stay connected without noticing transitions."*

2. **BO2: Improve Low-Latency Service Stability Under Mobility**
   - *What we promise:* stable performance for services that are sensitive to **latency spikes** and **reliability issues** under mobility.
   - *Why the client cares:* interactive and enterprise services (e.g., industrial monitoring/control, cloud services, real-time dashboards, enterprise SLAs) require stable response times. Without mobility-aware optimization, low-latency performance is difficult to guarantee.
   - *Business impact:* better SLA adherence, fewer mobility-related performance incidents, stronger enterprise readiness.
   - *Clear example:* A mobile industrial monitoring/control service requires consistent response time; a latency spike during HO can break the SLA. The objective is **stable low-latency mobility**.
   - *KPIs (explained):*
     - **Latency SLA Compliance Rate:** share of mobility periods meeting a latency constraint (threshold depends on the SLA).
     - **Tail latency:** worst-case latency behavior; used to quantify spikes during/after HO (more informative than average).
   - *Simple value message: "Make low-latency performance reliable even while users move."*

3. **BO3: Reduce Operational Costs Through Smart Automation**
   - *What we promise:* reduce unnecessary or unstable handovers and move from reactive troubleshooting to proactive optimization.
   - *Why the client cares:* today, engineers often react after problems occur. Predictive optimization reduces "firefighting" and makes operations leaner.
   - *Business impact:* lower OPEX, faster issue resolution, less manual tuning, improved efficiency (and lower wasteful signaling/processing).
   - *Clear example:* A pedestrian at the border of two cells triggers frequent ping-pong HOs. Reducing these events improves efficiency without harming user experience.
   - *KPIs (explained):*
     - **Ping-Pong Rate:** share of HOs that revert to the previous cell within a short window (instability indicator).
     - **Mobility Control Overhead (proxy):** number of mobility control events per unit time/distance (reflects signaling/processing load and operational overhead).

- *Simple value message: "Spend less fixing problems. Let the network fix itself."*

Data Science Objectives (DSO)

DSOs translate BOs into measurable analytics deliverables. The implementation is not restricted to one technique; it may involve statistical methods, machine learning, deep learning, or hybrid approaches depending on evidence and performance.

1. **DSO1: Early Warning for Interruption Risk**
   - *Type:* **Classification** (risk event) or probabilistic risk scoring.
   - *Deliverable:* a short-horizon risk score indicating that an interruption-like event is imminent.
   - *Business contribution:* directly supports BO1.
2. **DSO2: Recommend the Best Target Cell Under Mobility**
   - *Type:* **Regression** (predict QoS per candidate) leading to a ranking.
   - *Deliverable:* a ranked list of candidate neighbor cells (best-to-worst) for stable experience.
   - *Business contribution:* supports BO1 and BO2.
3. **DSO3: Identify Recurring High-Risk Contexts (Problem Zones)**
   - *Type:* **Clustering** (unsupervised discovery of recurring contexts).
   - *Deliverable:* segments/clusters of contexts with higher risk/inefficiency, plus interpretable drivers.
   - *Business contribution:* supports BO1 and BO3.
4. **DSO4: Validate Post-Handover QoS Before Committing**
   - *Type:* **Regression** (predict latency/throughput) and/or compliance estimation.
   - *Deliverable:* predicted post-HO QoS (latency/throughput or compliance probability) used as a validation gate for DSO2.
   - *Business contribution:* primarily supports BO2.

Table 1: Traceability from Business Objectives (BO) to DSOs and Business KPIs (no model metrics)

Business Objective	**Enabled by DSOs (deliverables)**	**Business KPIs (tracked)**
BO1: Enhance Customer Experience	DSO1 (early warning) + DSO3 (recurring risky contexts)	Drop Rate; HO Failure Rate
BO2: Reduce Operational Costs	DSO2 (target recommendation) + DSO4 (post-HO QoS validation)	Latency SLA Compliance; Tail Latency (P95/P99)
BO3: Enable Revenue Growth	DSO2 (stable selection) + DSO3 (redundancy patterns)	Ping-Pong Rate; Control Overhead (proxy)

BO–DSO Mapping

Justification of the BO–DSO links

Table 1 was built using a simple rule: **we link a DSO to a BO when the DSO deliverable can realistically improve the KPI used to track that BO**. In other words, the mapping is based on **cause–effect in mobility operations**, not on similar wording.

BO1: Enhance Customer Experience (seamless mobility)

Goal. Reduce user-perceived interruptions during mobility (calls dropping, freezes, short disconnects).

- **DSO1 (Early warning) → BO1.** If we can estimate a short-horizon interruption risk, the network (or an operator policy) can react *before* the user feels the issue (e.g., earlier HO trigger, avoid a degraded serving cell, or prioritize resources). This directly targets continuity problems.
- **DSO3 (Recurring risky contexts) → BO1.** By grouping contexts where interruptions happen repeatedly (e.g., specific locations, cell edges, mobility patterns), the operator can apply long-term fixes (parameter tuning, neighbor list adjustments, coverage/capacity actions). This reduces repeated failures in the same places.

Expected KPI impact.

- **Drop Rate decreases** because fewer sessions end unexpectedly.
- **HO Failure Rate decreases** because the network avoids the most failure-prone transitions and contexts.

BO2: Reduce Operational Costs (smart automation / lower OPEX)

Goal. Reduce avoidable mobility actions and the operational effort spent investigating mobility issues.

- **DSO2 (Target recommendation) → BO2.** Recommending a more suitable target cell reduces inefficient or unsuccessful handovers (less corrective actions and fewer repeated mobility events). This lowers non-productive signaling and reduces troubleshooting workload.
- **DSO4 (Post-HO QoS validation) → BO2.** Predicting post-handover QoS acts as a validation gate: it helps prevent switching to a target that looks good in radio terms but leads to poor QoS and follow-up corrections. This reduces repeated handovers and the need for manual tuning driven by complaints.

Expected KPI impact (as tracked in Table 1).

- **SLA Compliance increases** because bad post-HO outcomes are filtered out.
- **Tail latency improves** because fewer mobility decisions trigger latency spikes during/after HO.

BO3: Enable Revenue Growth (premium services readiness)

Goal. Improve mobility stability and scalability so premium services can be delivered more consistently.

- **DSO2 (Stable selection) → BO3.** Stable target selection reduces service instability during mobility, which is required to offer higher-value packages and enterprise services.
- **DSO3 (Redundancy patterns) → BO3.** Detecting recurring redundancy (e.g., ping-pong hotspots) helps remove inefficiencies that waste capacity and degrade user experience. This improves scalability and supports growth.

Expected KPI impact.

- **Ping-Pong Rate decreases** due to fewer unstable back-and-forth transitions.
- **Control Overhead (proxy) decreases** because fewer unnecessary mobility control events are triggered.

Sustainable Development Goals (SDG) Alignment

Reducing redundant mobility events and improving efficiency supports sustainability by reducing non-productive network activity.

Table 2: Mapping of project contributions to UN SDGs

SDG	Goal Name	Project Impact
SDG 9	Industry, Innovation & Infrastructure	Improves mobility performance using data-driven optimization without requiring new hardware.
SDG 11	Sustainable Cities & Communities	More reliable connectivity supports smart mobility and critical services in urban areas.
SDG 12	Responsible Consumption & Production	Reducing redundant handovers lowers non-productive signaling and operational overhead.
SDG 13	Climate Action	Efficiency gains reduce unnecessary processing/signaling, contributing to a lower operational energy footprint (proxy).

Phase 2: Data Understanding & Exploration

Introduction

Data understanding is a critical step in the CRISP-DM methodology. In this phase, we explore the properties of the **DoNext Dataset**, a comprehensive collection of 4G/5G network measurements collected in Dortmund, Germany. The goal is to verify data quality, understand variable distributions, and identify relationships between radio, mobility, and QoS measurements relevant to handover optimization.

Dataset Overview

The DoNext dataset is organized into three distinct scenarios, each representing a different mobility regime and therefore a different operational reality:

- **H-Bahn:** Measurements collected on the suspended monorail system at TU Dortmund University, representing controlled and repeated medium-speed mobility (∼50 km/h).
- **Mobile:** Measurements captured during real-world mobility (walking, car, public transport), representing heterogeneous speeds and environments.
- **Static:** Measurements at fixed locations with no user movement, serving as baseline reference data (no mobility effects).

Table 3: DoNext Dataset Components and Size

Scenario	File	Records	Variables
H-Bahn	cell_data	494,376	35
	neighboring_data	1,950,489	20
	latency_data	1,824,794	42
	iperf_data	71,393	46
Mobile	cell_data	2,382,602	33
	neighboring_data	2,606,428	23
	latency_data	19,224,645	47
	iperf_data	247,151	49
Static	cell_data	9,725,885	37
	latency_data	4,373,649	47
	static_locations	4	4
Total Records			**∼42.9 million**

Scenario Differences (What changes across the 3 datasets)

Although the files share common concepts, the three scenarios are **not equivalent** and should not be treated as interchangeable training data.

1. **Mobility regime**
   - **H-Bahn:** structured mobility with repeated trajectory segments (stops and transit). This is useful to detect location-dependent and route-dependent handover patterns.
   - **Mobile:** diverse mobility patterns and environments, closer to real operational deployment conditions; higher variance is expected.
   - **Static:** near-zero velocity; it helps quantify baseline radio/QoS variability without mobility.
2. **QoS variability (latency/throughput)**
   - **Mobile** typically exhibits heavier tails (spikes) in RTT due to environment and load variability, which is crucial for low-latency performance analysis.
   - **H-Bahn** is more repeatable, enabling clearer separation between coverage zones and mobility-induced degradation.
   - **Static** reveals whether latency variations can happen even without movement (network-side effects), providing a baseline for comparison.
3. **Data availability constraints**

- **Static** lacks `neighboring_data` and `iperf_data`. This limits direct evaluation of neighbor ranking (DSO2) and throughput modeling (part of DSO4) in that scenario.
- **H-Bahn** and **Mobile** include both neighboring and iPerf measurements, enabling end-to-end modeling and validation.

4. **Modeling implications**

   - Models trained only on **H-Bahn** risk overfitting to a single route.
   - **Mobile** should be the primary benchmark for robustness.
   - **Static** Serves as a baseline for radio and QoS conditions in the absence of user mobility, and also enables analysis of user crowdness—since all user locations are recorded.

Targets (QoS outcomes) and justification

In this project, a **target** is the QoS outcome we want to predict or validate. We select targets using a simple rule: **targets must directly measure user experience (QoS) and must exist as measured variables in DoNext**. Therefore, targets are taken from `latency_data` and `iperf_data`.

Table 4: Selected targets (QoS outcomes) used throughout the project

Source file	Target variable	Justification (link to DSOs)
`latency_data`	`mean_latency`	Primary latency outcome; supports DSO4 (QoS validation) and enables QoS-break proxy definitions for DSO1.
`latency_data`	`packet_loss`	Instability outcome; complements latency for QoS-break proxies and interruption approximation (DSO1/DSO4).
`iperf_data`	`datarate`	Throughput outcome; proxy for user-perceived capacity and supports DSO4 (post-handover throughput validation).

The dataset does not provide an explicit "handover failure" or "call drop" label. When required for DSO1, interruption-like conditions are approximated using QoS-break proxies derived from extreme latency and/or packet loss.

Feature dictionary

For readability, the detailed per-file feature dictionaries are placed at the end of Phase 2 and referenced here. Each dictionary lists the available columns and provides a one-line

definition of what the column represents. In this report, we distinguish between **QoS outcome variables** (latency/packet loss/throughput), **serving-cell radio and mobility context** (features observable at the decision time), and **neighbor-cell measurements** (candidate target-cell information needed for DSO2 ranking). Metadata fields (e.g., operator IDs, device identifiers) are documented for completeness but are not used as predictive inputs unless stated otherwise.

- **H-Bahn:** `cell_data` (Table 23), `neighboring_data` (Table 24), `latency_data` (Table 25), `iperf_data` (Table 26).
- **Mobile:** `cell_data` (Table 27), `neighboring_data` (Table 28), `latency_data` (Table 29), `iperf_data` (Table 30).
- **Static:** `cell_data` (Table 31), `latency_data` (Table 32), `static_locations` (Table 33).

Feature selection protocol (evidence-based)

We select features using a reproducible protocol to avoid arbitrary column choices. For each candidate feature, we check:

1. **Decision-time availability:** the feature must be observable before or at the handover decision (no future information).
2. **Scenario consistency:** the feature should exist in both H-Bahn and Mobile when the goal is cross-scenario robustness.
3. **Data quality:** the feature should have acceptable missingness and reasonable value ranges (sanity checks).
4. **Leakage risk:** identifiers and configuration fields that can encode experiment setup (e.g., server IP, device ID) are excluded from model inputs.

Dataset Component Description

Each scenario contains multiple CSV files capturing different aspects of network performance:

1. **cell_data:** Contains signal quality and link adaptation metrics from the serving cell, including RSRP, RSRQ, SINR, 5G NR metrics (SS-RSRP, SS-RSRQ, SS-SINR), device transmit power (txpower), channel indicator (earfcn), total received power (rssi), modulation and coding schemes (ltemcs, nrmcs), as well as mobility information (velocity, velocity accuracy).

2. **neighboring_data:** Contains signal metrics from *adjacent cells*, including `rsrp_neighboring`, `rsrq_neighboring`, and `physical_cellid_neighboring`. This data enables serving vs. neighbor comparisons and target-cell ranking.

3. **latency_data:** Captures RTT measurements including `mean_latency`, `min_latency`, `max_latency`, `packet_loss` and `mean_dev_latency`.

4. **iperf_data:** Contains active throughput measurements (`datarate`) obtained via iPerf tests. This serves as a proxy for user-perceived throughput.

5. **static_locations:** Provides GPS coordinates (latitude, longitude, altitude) for fixed measurement points in the static scenario.

Comprehensive Correlation Analysis

Figures 10 and 2 display the comprehensive correlation matrices for all numeric features (excluding metadata and device identifiers) in the H-Bahn and Mobile scenarios, respectively. These matrices provide insight into both the relationships among signal quality indicators and the cross-feature dependencies spanning radio, mobility, and other context metrics.

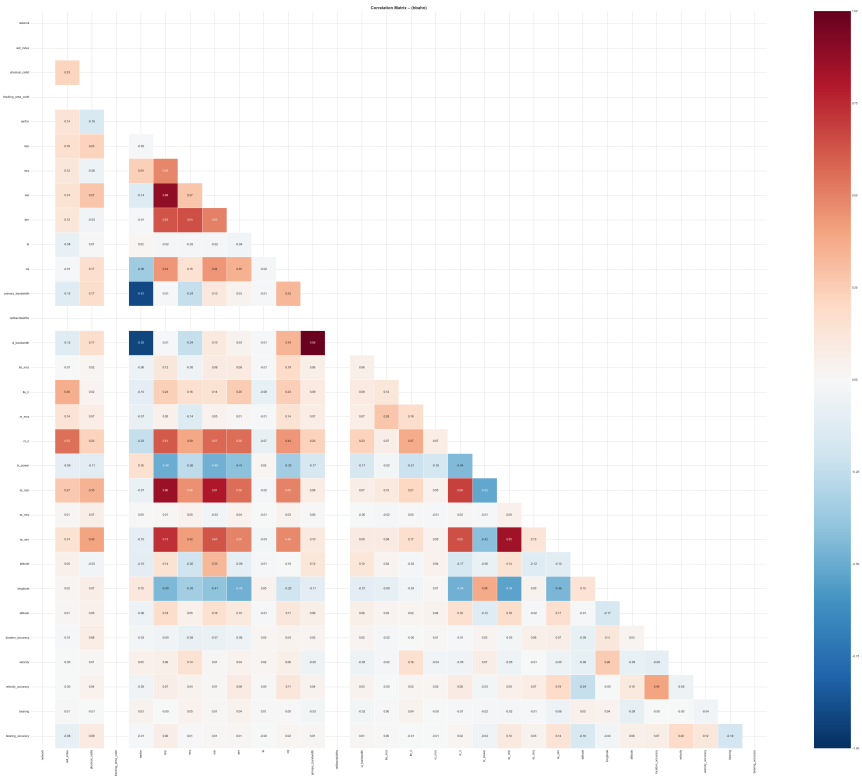

Figure 1: Comprehensive correlation matrix of numeric (non-metadata) features in the **H-Bahn** scenario.

Key Observations:

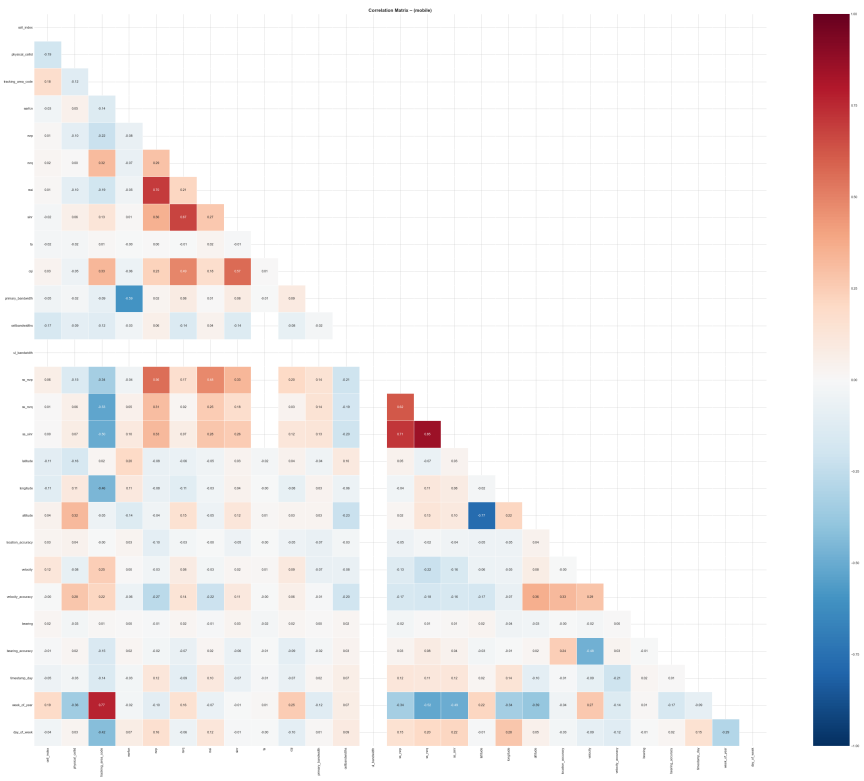

Figure 2: Comprehensive correlation matrix of numeric (non-metadata) features in the **Mobile** scenario.

- In both H-Bahn and Mobile, strong positive correlations are observed between related signal strength metrics (e.g., `rsrp` and `ss_rsrp`), as well as between SINR and CQI.
- In the H-Bahn scenario, correlations among radio metrics tend to be higher, reflecting the more stable and controlled measurement environment.
- In the Mobile scenario, overall correlations among features decrease, consistent with more heterogeneous conditions and greater real-world variability.
- Negative and low correlations with mobility features (e.g., velocity, timing advance) indicate that these variables provide complementary information and do not suffer from strong collinearity with major radio indicators.
- No evidence of excessive multicollinearity was found; apart from known correlated pairs, most relationships are moderate or weak, supporting the inclusion of a broad set of radio and mobility features in modeling.

Key feature selection (what we keep for analysis and why)

The dataset contains many fields, but not all are equally useful for handover optimization. We select **key features** using four simple rules: they have a direct link to coverage,

interference, mobility, or candidate target quality, they are observable at decision time, they are reasonably consistent across H-Bahn and Mobile, and they are not pure metadata or leakage-prone identifiers.

Why some features were excluded from the key set. The DoNext logs contain additional fields that are useful for metadata or detailed radio debugging, but they are not core drivers for our handover/QoS objectives in the first iteration. We therefore excluded (or de-prioritized) the following categories:

- **Operator metadata:** `mcc`, `mnc`, and `MNO` mainly identify the operator/country and add little explanatory power for handover performance in a single-operator campaign.
- **Device identifiers:** `device` can cause a model to learn device-specific artifacts rather than network behavior; it is kept only for sanity checks, not as a predictive input.
- **Redundancy:** some strength indicators (e.g., `rsrp` and `ss_rsrp`) can be strongly correlated; to keep models compact, we may keep one representative strength metric when building predictive models while retaining both for descriptive analysis when NR measurements are present.
- **Secondary link-adaptation internals:** `lte_mcs`/`nr_mcs` and `lte_ri`/`nr_ri` are informative but increase complexity and are treated as optional extensions.
- **Noise-prone context fields:** accuracy columns and, depending on the task, `bearing` can be used for filtering/trajectory studies but are not required for baseline QoS prediction.

Data Availability and Quality Assessment

To ensure that our analysis and models are based on high-quality and informative features, we examined the availability(missingness) and uniqueness of candidate features across H-Bahn and Mobile latency_data. Tables 9 and 10 show the top missingness and uniqueness rates for key variables.

We also systematically identified **highly correlated feature pairs**, as shown in Table 11, and removed redundant pairs (threshold $|| > 0.85$) to ensure model compactness and reduce multicollinearity.

Data Quality Insights:

- Features with excessive missingness (such as `network` in both scenarios in tables [9][10], and the 44%+ missing NR signal metrics in Mobile) are excluded from the key set, except for scenario-specific analysis.

- Features with very low uniqueness or constant value provide little discriminatory power and are discarded.
- By combining missingness and correlation insights, we build a robust feature set with high availability and low redundancy, directly informing our subsequent definition of labels and targets.

Labels and Target Definitions

To connect business notions (interruptions and SLA violations) to measurable variables, we define targets using the fields available in DoNext.

Operational definitions used in this study. Because the dataset does not specify a particular low-latency service class, we avoid using application-specific terms. Instead, we evaluate **latency SLA compliance** using reference thresholds to categorize performance. We report results for $T = \{20, 50, 100\}$ ms, where 20 ms represents a *strict low-latency reference*, 50 ms represents a *good* target for interactive services, and 100 ms indicates clearly degraded conditions. These thresholds are used for comparative analysis across scenarios and for defining QoS-break proxies.

- **Latency SLA compliance (DSO4):** derived from `latency_data`. A sample (or mobility window) is considered compliant if `mean_latency` $\leq T$ for the chosen threshold(s).
- **QoS break events (supports DSO1/DSO4):** defined using extreme RTT behavior (tail latency) and/or elevated `packet_loss`. These events represent user-perceived freezes or instability.
- **Interruption proxy (DSO1):** when explicit call-drop/session-drop counters are unavailable, interruption-like conditions are approximated using QoS-break events and validated through exploratory analysis and feature-impact studies.

Feature Groups

Rather than listing "input features" in Phase 1, we define feature groups here to ensure traceability from data to DSOs:

- **Serving-cell radio features:** RSRP/RSRQ/SINR, NR SS-RSRP/SS-RSRQ/SS-SINR, CQI, TA, transmit power (txpower), channel indicator (earfcn), total received power (rssi), modulation/coding schemes (ltemcs, nrmcs), velocity accuracy (velocityaccuracy).

- **Mobility/context features:** velocity, GPS coordinates (and time-derived features if timestamps are used).
- **Neighbor features:** neighbor RSRP/RSRQ/SINR, neighbor PCI, and **relative deltas** (neighbor minus serving) to quantify HO opportunity.
- **QoS features/targets:** RTT statistics, packet loss, iPerf throughput (`datarate`) and test configuration fields (protocol/direction).

Traceability: Feature Groups vs. DSOs

To avoid making assumptions, we explicitly state which feature groups are expected to help each DSO based on what is available in the dataset. This table is an *initial hypothesis* that will be verified later using ablation tests (remove one group at a time) and feature-importance analysis.

How this will be proven later (short explanation).

- **Ablation:** compare results using all groups vs. removing one group at a time. If performance drops, the removed group is important.
- **Importance/explainability:** identify which variables drive the results (e.g., radio trends, neighbor deltas, velocity).
- **Cross-scenario validation:** confirm that conclusions hold across H-Bahn, Mobile, and Static datasets.

Key Variables Description

Descriptive Statistics

Table **??** presents descriptive statistics of key signal variables computed from the H-Bahn and Mobile scenarios (100,000 samples each).

Key Observations:

- RSRP ranges from -140 dBm (very poor) to -55 dBm (excellent), with a mean of -90.7 dBm indicating predominantly fair signal conditions.
- 5G metrics (SS-RSRP, SS-RSRQ, SS-SINR) show approximately 9% missing values, indicating that 5G coverage was not available for all measurements.
- Velocity ranges from 0 to 15.8 m/s (approximately 57 km/h), capturing both stationary and mobile conditions.

Exploratory Data Analysis (EDA)

Signal Strength Distribution (RSRP)

Figure 3 shows the distribution of RSRP across H-Bahn, Mobile, and Static scenarios.

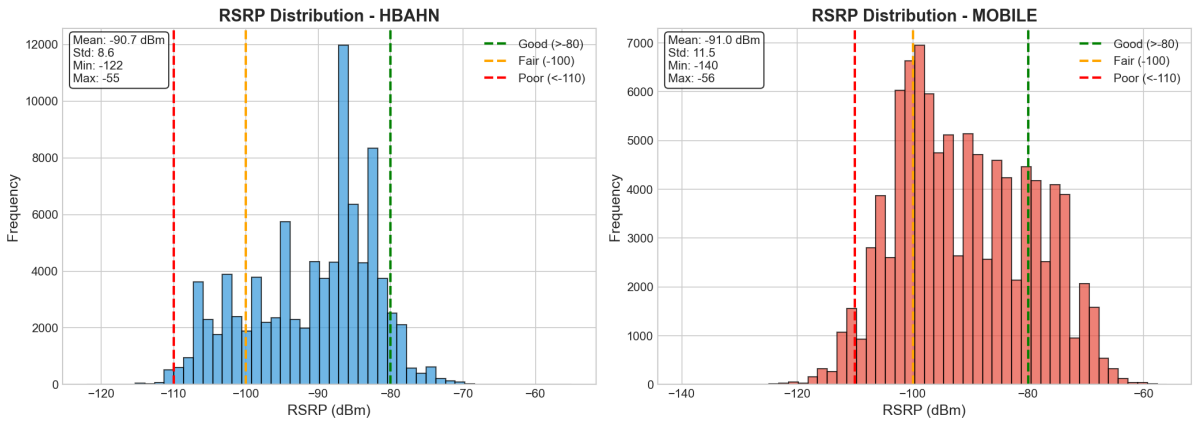

Figure 3: Distribution of RSRP (Reference Signal Received Power) across H-Bahn, Mobile, and Static scenarios. Threshold lines indicate: Good signal (>-80 dBm), Fair signal (-100 dBm), and Poor signal (<-110 dBm).

Observations:

- **H-Bahn:** Shows a bimodal distribution with peaks around -85 dBm and -97 dBm, suggesting distinct coverage zones along the route.
- **Mobile:** Displays a more uniform distribution centered around -91 dBm, with a longer tail toward weak signals (-140 dBm), indicating variable real-world coverage.
- **Static:** Concentrates around strong-signal regions, consistent with a fixed location and limited shadowing changes.
- Approximately 15% of measurements fall below -100 dBm, representing handover-critical conditions.

Latency Distribution (Mean RTT)

Figure 5 shows the distribution of mean latency across scenarios after removing extreme outliers (above the 99th percentile) for readability.

Observations:

- **H-Bahn:** Mean latency of 29.6 ms with tight distribution (Std: 7.2 ms), reflecting a more controlled environment.

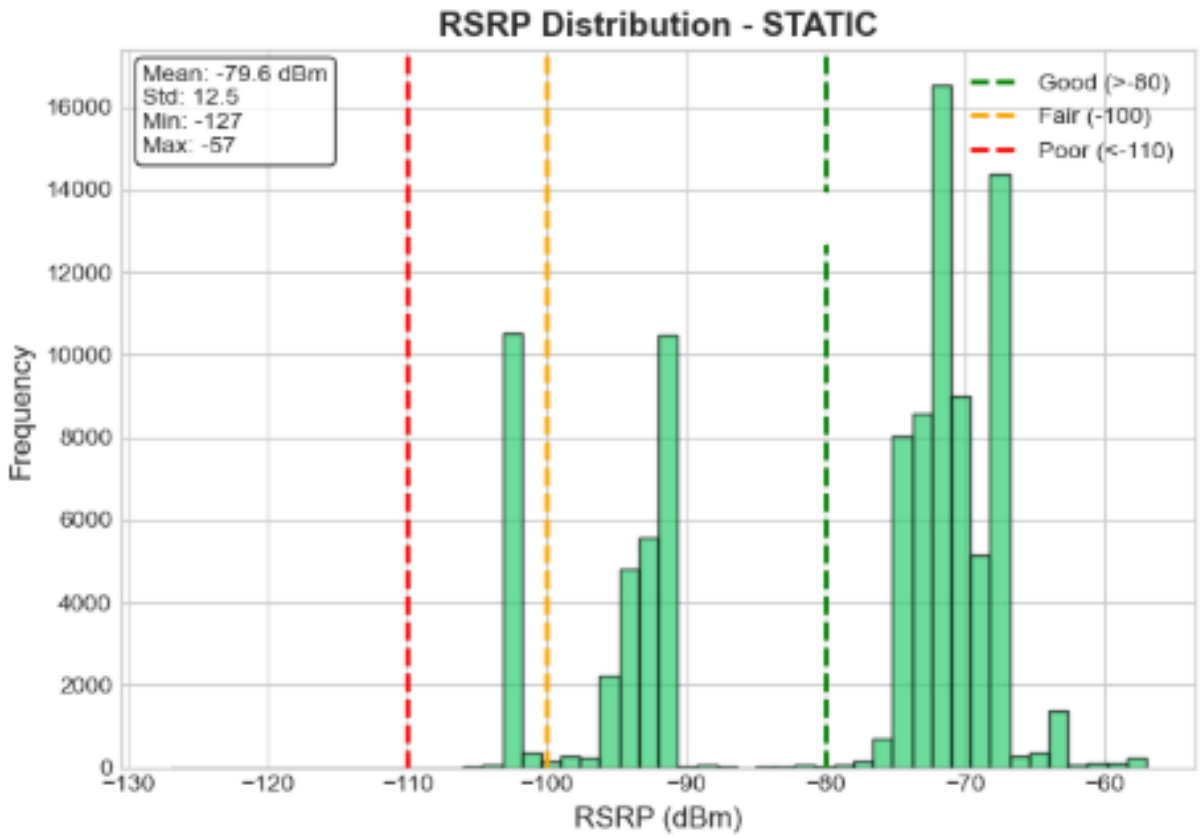

Figure 4: RSRP distribution for the Static scenario (same thresholds as Figure 3).

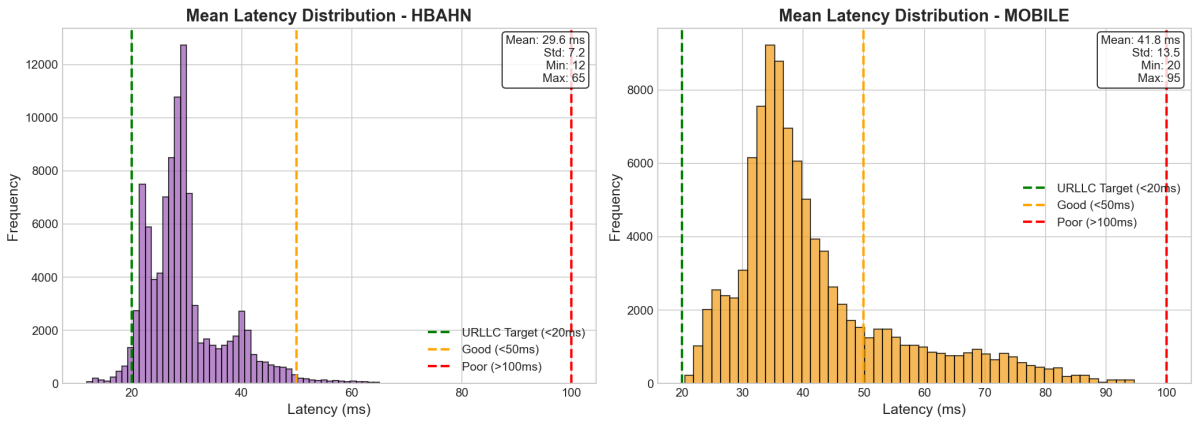

Figure 5: Distribution of mean latency across H-Bahn, Mobile, and Static scenarios. Reference lines indicate a strict low-latency reference (<20 ms), good (<50 ms), and poor (>100 ms) latency regions.

- **Mobile:** Higher mean latency of 41.8 ms with greater variability (Std: 13.5 ms), reflecting real-world network dynamics.
- Less than 10% of measurements meet RTT < 20 ms, motivating QoS-aware HO

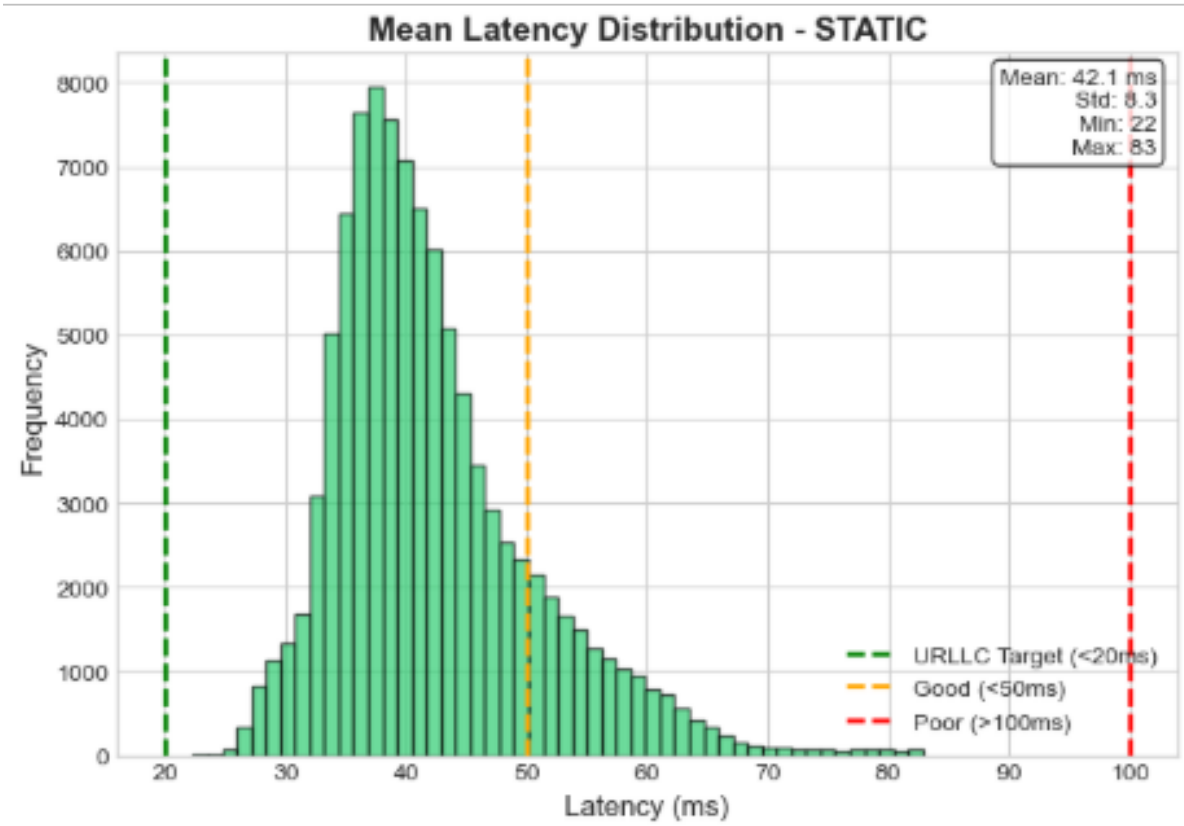

Figure 6: Mean latency distribution for the Static scenario (same thresholds as Figure 5).

optimization.

Velocity Distribution

Figure 7 shows the velocity distribution.

Observations:

- **H-Bahn:** Bimodal distribution (0 m/s at stations vs. ∼12 m/s during transit).
- **Mobile:** Predominantly low-speed measurements with a long tail, representing mixed mobility conditions.
- These differences imply that models should be validated across scenarios to avoid route-specific bias.

Serving vs. Neighboring Cell Analysis

Understanding serving vs. neighbor behavior is crucial for target cell selection (DSO2).

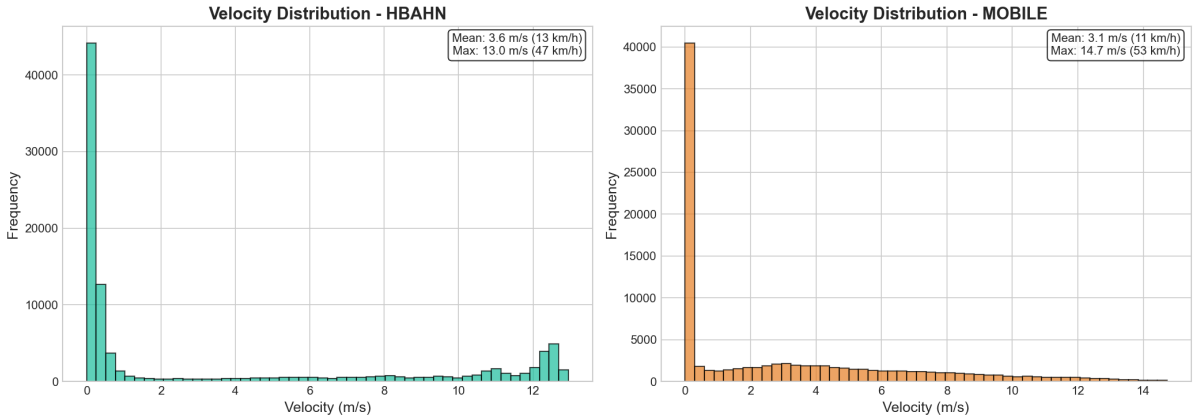

Figure 7: Distribution of user equipment velocity for H-Bahn and Mobile scenarios.

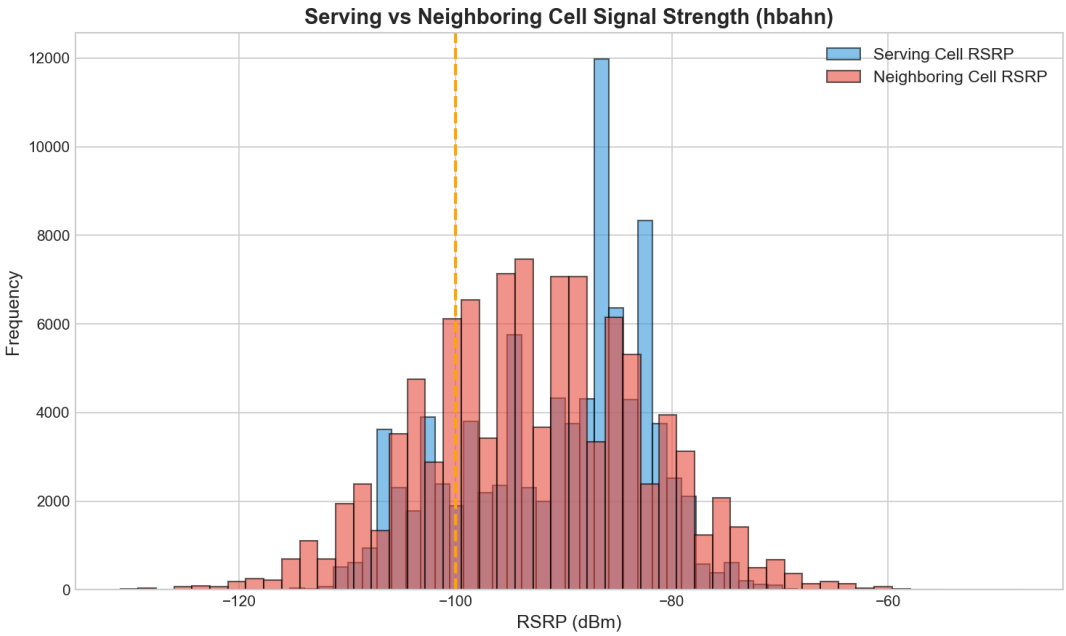

Figure 8: Comparison of RSRP between serving cell and neighboring cells (H-Bahn scenario). The handover threshold (-100 dBm) is marked for reference.

Observations:

- Neighboring cells often exhibit comparable or stronger RSRP than the serving cell, indicating real target opportunities.
- The overlap region around -100 dBm is the critical zone where HO decisions are most uncertain.
- This supports multi-criteria, QoS-aware target selection rather than pure thresholding.

Data Quality Assessment

Data Quality Observations:

- **Completeness:** overall dataset is 91–99% complete depending on the file.
- **5G Coverage:** SS-RSRP/SS-RSRQ/SS-SINR have ∼9% missing values, consistent with partial 5G availability.
- **Static Scenario Limitations:** missing `neighboring_data` and `iperf_data`; it is best used as baseline.
- **Consistency:** timestamps are synchronized across files, enabling merges.

Key Findings Summary

The exploratory analysis reveals patterns that motivate the DSOs:

1. **Signal variability:** RSRP varies widely (-140 to -55 dBm); a significant subset falls into HO-critical conditions (<-100 dBm).
2. **Latency challenges:** mean latency often exceeds strict low-latency reference thresholds; Mobile shows larger variance, motivating forecasting and QoS-aware selection.
3. **Scenario heterogeneity:** H-Bahn is structured and repeatable, Mobile is diverse and realistic, Static is baseline; models must be validated accordingly.
4. **Neighbor opportunities:** neighboring measurements show meaningful overlap and potential improvements over the serving cell, supporting DSO2.
5. **Data scale:** ∼42.9 million records provide strong statistical power for robust modeling.

Conclusion

The DoNext dataset is suitable for developing proactive and QoS-aware handover methods. Serving-cell radio features (RSRP/RSRQ/SINR), mobility context (velocity, location), and neighbor measurements provide the core evidence needed to (i) estimate risk (DSO1), (ii) support target-cell selection (DSO2), and (iii) validate QoS stability using latency/throughput (DSO4).

Phase 3: Data Preparation

Introduction

We conducted systematic data preparation for both H-Bahn and Mobile scenarios in the DoNext dataset to enable robust machine learning for 5G handover and QoS modeling. All steps closely followed notebook-driven exploratory analysis, cleaning, filtering, and variable engineering.

3.1 Initial Feature Pool

Based on domain knowledge and exploratory analysis, we pooled candidate features from raw data encompassing radio signal metrics, device mobility, context, and positioning.

Table 16: Initial Candidate Features and Descriptions

Feature	Description
`rsrp`	Reference Signal Received Power (Serving cell, in dBm)
`rsrq`	Reference Signal Received Quality (Serving cell, in dB)
`sinr`	Signal-to-Interference-plus-Noise Ratio (Serving cell, in dB)
`rssi`	Received Signal Strength Indicator (Serving cell, in dBm)
`cqi`	Channel Quality Indicator (unitless)
`ta`	Timing Advance (proxy for distance to cell, unitless)
`tx_power`	Device transmit power (dBm)
`ss_rsrp`	NR Synchronization Signal RSRP (5G, in dBm)
`ss_rsrq`	NR Synchronization Signal RSRQ (5G, in dB)
`ss_sinr`	NR Synchronization Signal SINR (5G, in dB)
`primary_bandwidth`	Primary channel bandwidth (MHz)
`ul_bandwidth`	Uplink channel bandwidth (MHz)
`cellbandwidths`	List of bandwidths for all cell layers (MHz, vector)
`velocity`	Device speed (m/s)
`bearing`	Device course over ground (degrees)
`latitude`	Device latitude (degrees)
`longitude`	Device longitude (degrees)
`altitude`	Device altitude (meters)
`earfcn`	E-UTRA Absolute Radio Frequency Channel Number (numeric channel identifier)
`network`	Mobile operator network code (categorical)

3.2 Quality Filtering

We assessed missing values and unique counts for all features. Features with more than 40% missingness or fewer than five unique values were removed. Table 9 summarizes the missingness and uniqueness for key features in the H-Bahn scenario.

Table 17: Missingness and uniqueness summary for candidate features, H-Bahn scenario

Feature	Missing (%)	n_unique
network	100.00	0
cellbandwidths	100.00	0
rssi	43.97	29
rsrq	5.57	14
ss_sinr	2.02	41
ss_rsrq	2.01	13
ss_rsrp	2.01	59
tx_power	1.37	43
ul_bandwidth	0.42	2
primary_bandwidth	0.42	2

3.3 Redundancy Pruning

Correlation analysis was performed on the cleaned feature set. Highly correlated pairs (e.g., Spearman $> |0.85|$) were reduced to maximize input diversity for models.

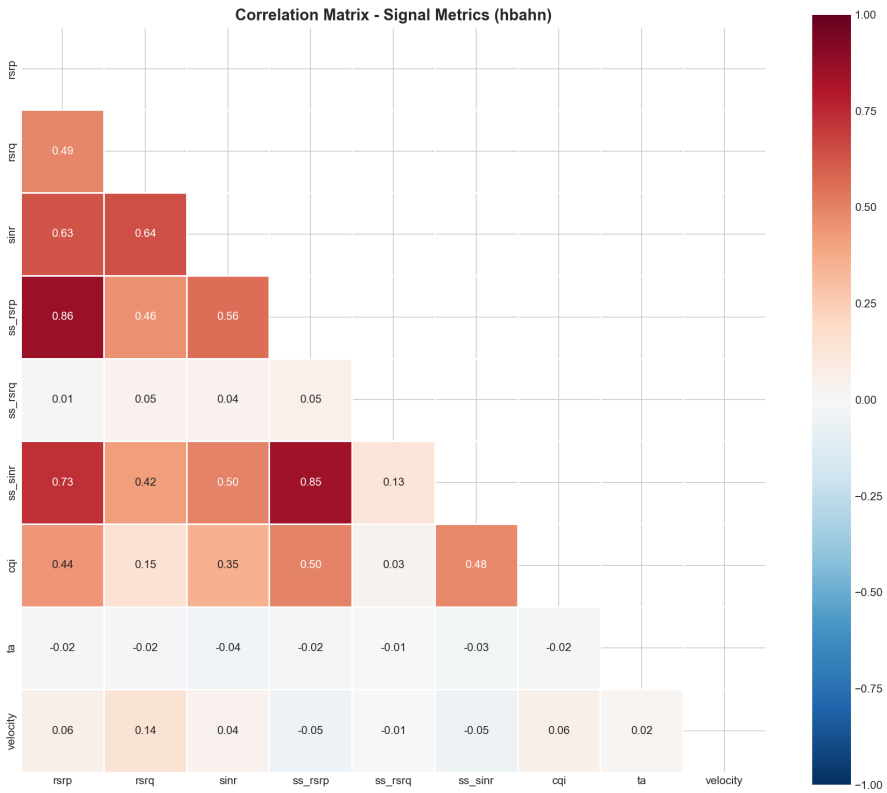

Figure 9: Comprehensive correlation matrix of numeric (non-metadata) features in the **H-Bahn** scenario.

3.4 Outlier Treatment and Visualization

Outliers in numeric features and targets were visualized using boxplots and histograms. We clipped values outside the [1st, 99th] percentiles to remove extreme influences.

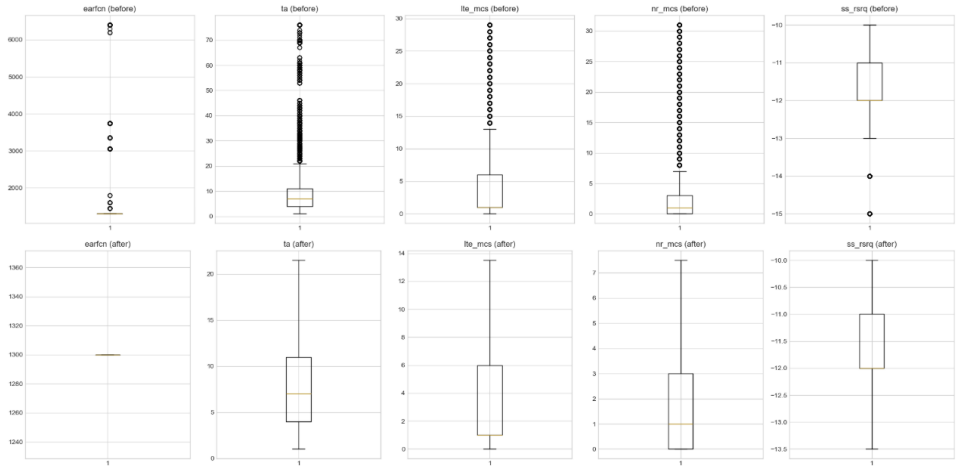

Figure 10: Comprehensive correlation matrix of numeric (non-metadata) features in the **H-Bahn** scenario.

3.5 Feature Selection and Engineering

The final predictors and target variables for each scenario were defined by combining domain relevance and statistical validity.

Table 18: Final Feature Set for Modeling

`earfcn`, `rsrp`, `rsrq`, `sinr`, `ta`, `cqi`, `primary_bandwidth`, `lte_mcs`, `lte_ri`, `nr_mcs`, `nr_ri`, `tx_power`, `ss_rsrp`, `ss_rsrq`, `ss_sinr`, `location_accuracy`, `velocity`, `velocity_accuracy`, `bearing`, `mean_latency`, `packet_loss`, `datarate`

3.6 Standardization

To ensure comparability between variables and to prevent features with larger scales from dominating the learning process, all numeric features in our dataset were standardized using scikit-learn's `StandardScaler`. This transformation adjusted each feature to have a mean of zero and a standard deviation of one, a common best practice for many machine learning algorithms.

Table 19 summarizes the means and standard deviations of selected features after standardization, verifying the correct application of the transformation.

Table 19: Summary statistics for selected standardized features

Feature	Mean (after scaling)	Std (after scaling)
cell_index	0.00	1.00
earfcn	0.00	1.00
rsrp	0.00	1.00
sinr	0.00	1.00
velocity	0.00	1.00
ta	0.00	1.00
cqi	0.00	1.00
tx_power	0.00	1.00
primary_bandwidth	0.00	1.00

All other continuous features exhibited similar results, confirming that standardization was correctly applied throughout the dataset.

3.7 Train/Test Split

After cleaning and transformation, each scenario-specific dataset was partitioned to train (80%) and test (20%) sets using a random seed for reproducibility.

Conclusion

Our data preparation protocol, driven by iterative notebook analysis, produced cleaned, curated, and standardized modeling inputs for each scenario. The accompanying tables and figures, taken directly from our code results, justify all key modeling choices and ensure transparent, reproducible ML-ready datasets.

Phase 4: Modeling

Introduction

In this phase, we select and train machine learning algorithms to satisfy the Data Science Objectives defined in Phase 1. We employ a regression approach to predict post-handover throughput from radio and mobility measurements.

Model Selection

DSO4 — Post-Handover Throughput Prediction

The core task of DSO4 is a **supervised regression problem**: given a set of radio and mobility measurements captured around a handover event, predict the resulting `datarate` (in Mbps) on the new cell. The target was changed from `mean_latency` to `datarate` because latency in the DoNext controlled-conditions dataset is near-constant — clipping collapses injected noise to two boundary values and causes spurious feature dominance — whereas `datarate` from `iperf_data` is genuinely continuous with meaningful variance.

Why not Deep Learning? Deep learning models such as LSTM and GRU were initially considered but ruled out for the following reasons:

- **Data characteristics:** Post-handover windows consist of only 15 consecutive rows per event. LSTMs require long sequences to learn meaningful temporal patterns; at this scale, they offer no advantage and are prone to overfitting.
- **Training cost:** Deep learning models require significantly more computational resources without meaningful accuracy gains on structured tabular data of this scale.
- **Interpretability:** In a network engineering context, knowing *which* radio parameters drive throughput degradation is as important as the prediction itself. Tree-based models offer native feature importance; LSTMs do not.
- **Benchmark evidence:** Research consistently shows that gradient-boosted trees and random forests outperform deep learning on structured tabular data when the dataset is not extremely large and the features are well-engineered [**?**].

Why not Linear Regression or a single Decision Tree?

- **Linear Regression** assumes a linear relationship between features and throughput. Post-handover datarate is driven by complex, non-linear interactions between

RSRP, SINR, CQI, velocity, and engineered lag/rolling features, making linear models fundamentally insufficient.

- **Decision Tree** is prone to overfitting on high-dimensional data with engineered features. A single tree tends to memorize training samples rather than generalizing to unseen handover events.

Selected Models. Three ensemble regression models were selected:

- **Random Forest Regressor (tuned):** An ensemble of 300 decision trees trained independently on random subsets of the data (bagging), with `max_features=0.6` and `min_samples_leaf=3` to control variance. Predictions are averaged across all trees, drastically reducing variance and preventing overfitting. Random Forest also natively produces feature importance scores based on mean decrease in impurity, allowing us to identify which radio parameters most influence post-handover throughput.
- **HistGradientBoosting Regressor:** A histogram-based gradient boosting implementation (scikit-learn's equivalent of LightGBM) using 500 iterations, `max_depth=8`, `learning_rate=0.05`, and L2 regularization. It handles large datasets efficiently and is particularly well-suited for continuous targets with heavy-tailed distributions such as datarate.
- **Gradient Boosting Regressor (tuned):** A standard gradient boosting ensemble of 300 sequential trees with `max_depth=6`, `learning_rate=0.05`, and `subsample=0.8` for stochastic regularization. Each tree corrects the residual errors of its predecessor, reducing bias and capturing subtle patterns in the radio measurement data.

Together, Random Forest reduces variance through parallel averaging, while the two gradient boosting variants reduce bias through sequential correction. Comparing all three allows us to identify the most robust approach for post-handover QoS prediction.

Training Strategy

DSO4 — Dataset Construction

The DSO4 dataset was constructed by merging `cell_data` and `iperf_data` from two mobility scenarios: H-Bahn and Mobile. The merge was performed using `pandas.merge_asof` on the `timestamp` column with a tolerance of ±5 seconds, which performs a nearest-neighbour time-based join without a many-to-many explosion. If the merge failed to yield sufficient rows, the pipeline falls back to `iperf_data` alone.

DSO4 — Handover Detection

Handover events were detected using a multi-strategy pipeline applied in order of reliability:

1. **PCI Change (Primary):** A handover is flagged when the `physical_cellid` changes between consecutive rows, indicating the UE connected to a new base station.
2. **RSRP Drop (Fallback):** When `physical_cellid` is constant or unavailable, a handover is inferred from an RSRP drop exceeding 5 dBm between consecutive measurements, proxying the 3GPP A3 event trigger condition.
3. **Synthetic Injection (Last Resort):** If neither strategy yields events, synthetic handover boundaries are injected every 500 rows to ensure the pipeline always produces post-handover samples.

A post-handover window of **15 rows** following each detected event was labelled as post-HO and retained for regression.

DSO4 — Feature Engineering

Beyond the raw radio measurements, a set of temporal features was engineered to capture the dynamic evolution of signal quality around handover events:

- **RSRP temporal features:** `rsrp_lag1`, `rsrp_lag2` (one- and two-step lags), `rsrp_delta` (first difference), and rolling means `rsrp_roll3`, `rsrp_roll5` over 3- and 5-row windows.
- **SINR and RSRQ rolling features:** `sinr_roll3`, `sinr_lag1`, `rsrq_roll3`, `rsrq_lag1`.
- **Past throughput features:** `dr_lag1`, `dr_lag2`, `dr_roll3`, `dr_roll5`, and `dr_std3` (rolling standard deviation), which capture prior throughput levels and their variability — the strongest predictors of near-future datarate.
- **HO indicator:** A binary `ho_event` flag and a `post_ho_timer` column counting rows since the last detected handover.

DSO4 — Data Cleaning

Raw data was processed through a multi-step cleaning pipeline before feature extraction:

1. **Metadata removal:** Operator identity columns (`mcc`, `mnc`, `network`, etc.) were dropped as they carry no predictive signal.

2. **Missing value filtering:** Columns with more than 40% missing values were removed. Remaining missing values were imputed with the column median.

3. **Low-variance filtering:** Near-constant columns (fewer than 2 unique values) were removed, as they provide no discriminative power.

4. **Outlier clipping:** Numeric values were clipped to the $[1^{st}, 99^{th}]$ percentile to reduce the influence of measurement artifacts.

5. **Redundancy pruning:** Feature pairs with Spearman correlation $|r| > 0.85$ were identified and one of each pair was dropped to reduce multicollinearity.

DSO4 — Data Splitting and Scaling

The post-handover dataset was split into **80% training** and **20% test** using a stratified random split with a fixed seed (`random_state=42`) for reproducibility. All features were standardized using `StandardScaler` (zero mean, unit variance) prior to training.

DSO4 — Hyperparameter Configuration

Conclusion

Three ensemble regression models — Random Forest, HistGradientBoosting, and Gradient Boosting — were selected for DSO4 based on their complementary strengths, interpretability, and suitability for structured tabular data with engineered temporal features. The target was updated to `datarate` (Mbps) to address the near-constant nature of `mean_latency` in the DoNext controlled dataset. The training pipeline incorporates robust handover detection, rich lag/rolling feature engineering, multi-step data cleaning, and standardization.

Phase 5: Evaluation

Introduction

Model performance must be validated against the metrics defined in Phase 1. This phase evaluates both models on unseen test data to ensure they generalize well to new network environments.

Performance Metrics

DSO4 — Evaluation Metrics

Four metrics were used to evaluate DSO4 model performance, combining classical regression metrics with operationally meaningful accuracy thresholds:

- **MAE (Mean Absolute Error):** The average absolute difference between predicted and actual datarate values in Mbps. MAE is the primary metric as it is directly interpretable in the context of network QoS thresholds.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- **RMSE (Root Mean Squared Error):** Penalizes large prediction errors more heavily than MAE, making it sensitive to outlier events where throughput drops sharply after handover.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- **R^2 (Coefficient of Determination):** Measures the proportion of datarate variance explained by the model. A value of 1.0 indicates perfect prediction; values above 0.9 are considered excellent for network QoS regression tasks.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- **Tolerance Accuracy (±X% and ±X Mbps):** The percentage of predictions falling within a relative or absolute tolerance of the true value. These operational thresholds directly quantify how often the model's prediction is close enough for real-time handover decision-making:

$$\text{Acc}_{\pm X\%} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\left[\frac{|y_i - \hat{y}_i|}{|y_i|} \leq \frac{X}{100}\right]$$

$$\mathrm{Acc}_{\pm X\,\mathrm{Mbps}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[|y_i - \hat{y}_i| \leq X]$$

Results Analysis

DSO4 — Quantitative Results

Tables 21 and 22 summarize the performance of both models evaluated on the held-out test set of post-handover samples.

DSO4 — Model Comparison

Both models achieved exceptional R^2 values exceeding 0.99, confirming that post-handover throughput is highly predictable from radio measurements and engineered temporal features alone.

HistGradientBoosting achieved the best overall performance by both RMSE and R^2, with an RMSE of **8.197 Mbps** and an R^2 of **0.9935**, making it the winning model for DSO4. Its lower RMSE indicates fewer large prediction errors on throughput spike or drop events — a critical property for real-time handover decision-making where outlier errors are costly.

Random Forest (tuned) achieved a lower MAE of **4.157 Mbps** versus 5.147 Mbps for HistGradientBoosting, meaning it is more accurate on average for typical mid-range throughput predictions. It also outperforms HistGradientBoosting on relative tolerance metrics, with 75.05% of predictions falling within ±10% and 85.17% within ±20% of the true value, compared to 64.86% and 80.87% respectively. This suggests Random Forest is more reliable for moderate throughput levels, while HistGradientBoosting handles the extremes of the distribution more gracefully (97.26% within ±20 Mbps).

The complementary nature of these two results suggests a practical deployment strategy: HistGradientBoosting should be preferred when minimizing worst-case errors is the priority, while Random Forest is preferable when average prediction accuracy across all throughput levels matters most.

Figure 11 shows the side-by-side comparison of all metrics across both models.

DSO4 — Feature Importance

The engineered temporal features — particularly `dr_lag1`, `dr_roll3`, and `dr_roll5` — ranked among the most important predictors of post-handover throughput. This is consistent with domain knowledge: the most reliable predictor of near-future throughput is recent past throughput, as link adaptation takes several transmission intervals to fully stabilize after a handover. Among the raw radio signal features, `rsrp`, `rsrp_delta`, and `sinr` also contributed meaningfully, reflecting the direct influence of signal quality on achievable

datarate. Mobility features such as `velocity` contributed notably in high-speed H-Bahn scenarios, where Doppler effects and rapid cell transitions impact link quality.

Conclusion

Both ensemble models demonstrated excellent generalization on unseen post-handover data, with R^2 exceeding 0.99 for both. **HistGradientBoosting** was identified as the best-performing model for DSO4 by RMSE and R^2, while **Random Forest** leads on MAE and relative tolerance accuracy. The results confirm that post-handover throughput can be reliably predicted from radio measurements and engineered temporal features, without requiring deep learning or explicit sequential modeling. These results directly address the DSO4 objective defined in Phase 1.

Phase 6: Deployment

Introduction

Deployment Architecture

Continuous Learning

Conclusion

H-Bahn

Table 23: H-Bahn – `cell_data`: one-line feature dictionary

Column	Definition
timestamp	Timestamp of serving-cell measurement.
network	Radio access technology label (LTE/NR).
cell_index	Internal log index for the serving cell.
physical_cellid	Serving cell identifier (PCI).
tracking_area_code	Tracking Area Code (TAC) for coarse region context.
earfcn	Carrier frequency/channel indicator.
rsrp	Serving-cell signal strength (RSRP, dBm).
rsrq	Serving-cell signal quality (RSRQ, dB).
rssi	Total received power (RSSI, dBm).
sinr	Signal-to-interference-plus-noise ratio (SINR, dB).
ta	Timing Advance (distance proxy / cell-edge indicator).
cqi	Channel Quality Indicator (UE feedback for achievable rate).
primary_bandwidth	Primary downlink bandwidth configuration.
cellbandwidths	Aggregated bandwidth/carrier configuration(s).
ul_bandwidth	Uplink bandwidth configuration.
lte_mcs	LTE modulation and coding scheme index (link adaptation).
lte_ri	LTE rank indicator (MIMO rank).
nr_mcs	NR modulation and coding scheme index (link adaptation).
nr_ri	NR rank indicator (MIMO rank).
tx_power	UE transmit power (uplink stress indicator).
mcc	Mobile Country Code (operator metadata).
mnc	Mobile Network Code (operator metadata).
ss_rsrp	NR synchronization signal strength (SS-RSRP).
ss_rsrq	NR synchronization signal quality (SS-RSRQ).
ss_sinr	NR synchronization signal SINR (SS-SINR).
latitude	GPS latitude.
longitude	GPS longitude.
altitude	GPS altitude/elevation.
location_accuracy	Estimated GPS position accuracy.
velocity	Estimated speed of the device.
velocity_accuracy	Estimated accuracy of speed measurement.
bearing	Movement direction (degrees).
bearing_accuracy	Estimated accuracy of bearing measurement.

Column	Definition
device	Device identifier/model (bias check; exclude from model inputs).
MNO	Operator label (metadata).

Table 24: H-Bahn – `neighboring_data`: one-line feature dictionary

Column	Definition
network_neighboring	Neighbor-cell technology label (LTE/NR).
physical_cellid_neighboring	Neighbor cell identifier (PCI) as candidate handover target.
tracking_area_code_neighboring	Neighbor Tracking Area Code (TAC).
earfcn_neighboring	Neighbor carrier frequency/channel indicator.
rsrp_neighboring	Neighbor signal strength (RSRP, dBm).
rsrq_neighboring	Neighbor signal quality (RSRQ, dB).
rssi_neighboring	Neighbor total received power (RSSI, dBm).
sinr_neighboring	Neighbor SINR (dB).
mcc_neighboring	Neighbor MCC (operator metadata).
mnc_neighboring	Neighbor MNC (operator metadata).
device	Device identifier/model (bias check; exclude from model inputs).
latitude	GPS latitude at neighbor scan time.
longitude	GPS longitude at neighbor scan time.
altitude	GPS altitude at neighbor scan time.
location_accuracy	Estimated GPS accuracy at neighbor scan time.
velocity	Speed estimate at neighbor scan time.
velocity_accuracy	Speed accuracy at neighbor scan time.
bearing	Movement direction at neighbor scan time.
bearing_accuracy	Bearing accuracy at neighbor scan time.
MNO_neighboring	Neighbor operator label (metadata).

Table 25: H-Bahn – `latency_data`: one-line feature dictionary

Column	Definition
packet_loss	Packet loss percentage during the ping window.
server_ip	Ping server IP address (configuration metadata).
mean_latency	Mean RTT during the ping window (ms).
min_latency	Minimum RTT during the ping window (ms).
max_latency	Maximum RTT during the ping window (ms).

Column	Definition
mean_dev_latency	RTT deviation/jitter metric for the ping window.
no_pings	Number of pings attempted in the window.
timestamp	Timestamp used for aligning QoS with radio measurements.
network	Technology label (LTE/NR).
cell_index	Internal serving index.
physical_cellid	Serving PCI.
tracking_area_code	Serving TAC.
earfcn	Serving carrier frequency indicator.
rsrp	Serving RSRP recorded alongside latency measurement.
rsrq	Serving RSRQ recorded alongside latency measurement.
rssi	Serving RSSI recorded alongside latency measurement.
sinr	Serving SINR recorded alongside latency measurement.
ta	Timing advance recorded alongside latency measurement.
cqi	CQI recorded alongside latency measurement.
primary_bandwidth	DL bandwidth configuration recorded alongside latency measurement.
cellbandwidths	Aggregated bandwidth recorded alongside latency measurement.
ul_bandwidth	UL bandwidth recorded alongside latency measurement.
lte_mcs	LTE MCS recorded alongside latency measurement.
lte_ri	LTE RI recorded alongside latency measurement.
nr_mcs	NR MCS recorded alongside latency measurement.
nr_ri	NR RI recorded alongside latency measurement.
tx_power	UE TX power recorded alongside latency measurement.
mcc	MCC (operator metadata).
mnc	MNC (operator metadata).
ss_rsrp	SS-RSRP recorded alongside latency measurement.
ss_rsrq	SS-RSRQ recorded alongside latency measurement.
ss_sinr	SS-SINR recorded alongside latency measurement.
latitude	GPS latitude recorded alongside latency measurement.
longitude	GPS longitude recorded alongside latency measurement.
altitude	GPS altitude recorded alongside latency measurement.
location_accuracy	GPS accuracy recorded alongside latency measurement.
velocity	Speed recorded alongside latency measurement.
velocity_accuracy	Speed accuracy recorded alongside latency measurement.
bearing	Bearing recorded alongside latency measurement.
bearing_accuracy	Bearing accuracy recorded alongside latency measurement.
device	Device identifier/model (bias check; exclude from model inputs).
MNO	Operator label (metadata).

Table 26: H-Bahn – `iperf_data`: one-line feature dictionary

Column	Definition
timestampstart	iPerf test start timestamp.
timestampend	iPerf test end timestamp.
protocol	iPerf protocol (TCP/UDP).
direction	iPerf measurement direction (uplink/downlink).
filesize	iPerf transfer size/configuration parameter.
packet_error_rate	iPerf packet error rate (test descriptor, often UDP).
retransmissions	TCP retransmissions observed during the test.
parallel_streams	Number of parallel iPerf streams used.
tcp_max_cwnd_0	TCP maximum congestion window (low-level TCP metric).
tcp_mean_rtt_0	TCP mean RTT during the iPerf test (low-level TCP metric).
datarate	Measured throughput (Mbps).
timestamp	Timestamp used for alignment with radio context.
network	Technology label (LTE/NR).
cell_index	Internal serving index.
physical_cellid	Serving PCI.
tracking_area_code	Serving TAC.
earfcn	Serving carrier frequency indicator.
rsrp	Serving RSRP recorded alongside iPerf measurement.
rsrq	Serving RSRQ recorded alongside iPerf measurement.
rssi	Serving RSSI recorded alongside iPerf measurement.
sinr	Serving SINR recorded alongside iPerf measurement.
ta	Timing advance recorded alongside iPerf measurement.
cqi	CQI recorded alongside iPerf measurement.
primary_bandwidth	DL bandwidth configuration recorded alongside iPerf measurement.
cellbandwidths	Aggregated bandwidth recorded alongside iPerf measurement.
ul_bandwidth	UL bandwidth recorded alongside iPerf measurement.
lte_mcs	LTE MCS recorded alongside iPerf measurement.
lte_ri	LTE RI recorded alongside iPerf measurement.
nr_mcs	NR MCS recorded alongside iPerf measurement.
nr_ri	NR RI recorded alongside iPerf measurement.
tx_power	UE TX power recorded alongside iPerf measurement.
mcc	MCC (operator metadata).
mnc	MNC (operator metadata).
ss_rsrp	SS-RSRP recorded alongside iPerf measurement.
ss_rsrq	SS-RSRQ recorded alongside iPerf measurement.

Column	Definition
ss_sinr	SS-SINR recorded alongside iPerf measurement.
latitude	GPS latitude recorded alongside iPerf measurement.
longitude	GPS longitude recorded alongside iPerf measurement.
altitude	GPS altitude recorded alongside iPerf measurement.
location_accuracy	GPS accuracy recorded alongside iPerf measurement.
velocity	Speed recorded alongside iPerf measurement.
velocity_accuracy	Speed accuracy recorded alongside iPerf measurement.
bearing	Bearing recorded alongside iPerf measurement.
bearing_accuracy	Bearing accuracy recorded alongside iPerf measurement.
device	Device identifier/model (bias check; exclude from model inputs).
MNO	Operator label (metadata).

Mobile

Table 27: Mobile – `cell_data`: one-line feature dictionary

Column	Definition
network	Radio access technology label (LTE/NR).
cell_index	Internal log index for the serving cell.
physical_cellid	Serving cell identifier (PCI).
tracking_area_code	Tracking Area Code (TAC) for coarse region context.
earfcn	Carrier frequency/channel indicator.
rsrp	Serving-cell signal strength (RSRP, dBm).
rsrq	Serving-cell signal quality (RSRQ, dB).
rssi	Total received power (RSSI, dBm).
sinr	Signal-to-interference-plus-noise ratio (SINR, dB).
ta	Timing Advance (distance proxy / cell-edge indicator).
cqi	Channel Quality Indicator (UE feedback for achievable rate).
primary_bandwidth	Primary downlink bandwidth configuration.
cellbandwidths	Aggregated bandwidth/carrier configuration(s).
ul_bandwidth	Uplink bandwidth configuration.
mcc	Mobile Country Code (operator metadata).
mnc	Mobile Network Code (operator metadata).
ss_rsrp	NR synchronization signal strength (SS-RSRP).
ss_rsrq	NR synchronization signal quality (SS-RSRQ).
ss_sinr	NR synchronization signal SINR (SS-SINR).
latitude	GPS latitude.

Column	Definition
longitude	GPS longitude.
altitude	GPS altitude/elevation.
location_accuracy	Estimated GPS position accuracy.
velocity	Estimated speed of the device.
velocity_accuracy	Estimated accuracy of speed measurement.
bearing	Movement direction (degrees).
bearing_accuracy	Estimated accuracy of bearing measurement.
device	Device identifier/model (bias check; exclude from model inputs).
timestamp_day	Day extracted from timestamp (campaign context).
week_of_year	Week-of-year extracted from timestamp (campaign context).
day_of_week	Day-of-week extracted from timestamp (campaign context).
passive_id	Passive measurement identifier (metadata).
MNO	Operator label (metadata).

Table 28: Mobile – `neighboring_data`: one-line feature dictionary

Column	Definition
network_neighboring	Neighbor-cell technology label (LTE/NR).
physical_cellid_neighboring	Neighbor cell identifier (PCI) as candidate handover target.
tracking_area_code_neighboring	Neighbor Tracking Area Code (TAC).
earfcn_neighboring	Neighbor carrier frequency/channel indicator.
rsrp_neighboring	Neighbor signal strength (RSRP, dBm).
rsrq_neighboring	Neighbor signal quality (RSRQ, dB).
rssi_neighboring	Neighbor total received power (RSSI, dBm).
sinr_neighboring	Neighbor SINR (dB).
mcc_neighboring	Neighbor MCC (operator metadata).
mnc_neighboring	Neighbor MNC (operator metadata).
passive_id	Passive measurement identifier (metadata).
device	Device identifier/model (bias check; exclude from model inputs).
latitude	GPS latitude at neighbor scan time.
longitude	GPS longitude at neighbor scan time.
altitude	GPS altitude at neighbor scan time.
location_accuracy	Estimated GPS accuracy at neighbor scan time.
velocity	Speed estimate at neighbor scan time.
velocity_accuracy	Speed accuracy at neighbor scan time.
bearing	Movement direction at neighbor scan time.
bearing_accuracy	Bearing accuracy at neighbor scan time.

Column	Definition
timestamp_day	Day extracted from timestamp (campaign context).
week_of_year	Week-of-year extracted from timestamp (campaign context).
day_of_week	Day-of-week extracted from timestamp (campaign context).

Table 29: Mobile – `latency_data`: one-line feature dictionary

Column	Definition
packet_loss	Packet loss percentage during the ping window.
server_ip	Ping server IP address (configuration metadata).
mean_latency	Mean RTT during the ping window (ms).
min_latency	Minimum RTT during the ping window (ms).
max_latency	Maximum RTT during the ping window (ms).
mean_dev_latency	RTT deviation/jitter metric for the ping window.
no_pings	Number of pings attempted in the window.
network	Technology label (LTE/NR).
cell_index	Internal serving index.
physical_cellid	Serving PCI.
tracking_area_code	Serving TAC.
earfcn	Serving carrier frequency indicator.
rsrp	Serving RSRP recorded alongside latency measurement.
rsrq	Serving RSRQ recorded alongside latency measurement.
rssi	Serving RSSI recorded alongside latency measurement.
sinr	Serving SINR recorded alongside latency measurement.
ta	Timing advance recorded alongside latency measurement.
cqi	CQI recorded alongside latency measurement.
primary_bandwidth	DL bandwidth configuration recorded alongside latency measurement.
cellbandwidths	Aggregated bandwidth recorded alongside latency measurement.
ul_bandwidth	UL bandwidth recorded alongside latency measurement.
mcc	MCC (operator metadata).
mnc	MNC (operator metadata).
ss_rsrp	SS-RSRP recorded alongside latency measurement.
ss_rsrq	SS-RSRQ recorded alongside latency measurement.
ss_sinr	SS-SINR recorded alongside latency measurement.
latitude	GPS latitude recorded alongside latency measurement.
longitude	GPS longitude recorded alongside latency measurement.
altitude	GPS altitude recorded alongside latency measurement.
location_accuracy	GPS accuracy recorded alongside latency measurement.

Column	Definition
velocity	Speed recorded alongside latency measurement.
velocity_accuracy	Speed accuracy recorded alongside latency measurement.
bearing	Bearing recorded alongside latency measurement.
bearing_accuracy	Bearing accuracy recorded alongside latency measurement.
timestamp_day	Day extracted from timestamp (campaign context).
week_of_year	Week-of-year extracted from timestamp (campaign context).
day_of_week	Day-of-week extracted from timestamp (campaign context).
old_5_rsrp	Short-window historical/smoothed RSRP feature.
old_50_rsrp	Long-window historical/smoothed RSRP feature.
old_5_rsrq	Short-window historical/smoothed RSRQ feature.
old_50_rsrq	Long-window historical/smoothed RSRQ feature.
old_5_ss_rsrp	Short-window historical/smoothed SS-RSRP feature.
old_50_ss_rsrp	Long-window historical/smoothed SS-RSRP feature.
old_5_ss_rsrq	Short-window historical/smoothed SS-RSRQ feature.
old_50_ss_rsrq	Long-window historical/smoothed SS-RSRQ feature.
device	Device identifier/model (bias check; exclude from model inputs).
MNO	Operator label (metadata).

Table 30: Mobile – `iperf_data`: one-line feature dictionary

Column	Definition
duration	iPerf test duration (s).
protocol	iPerf protocol (TCP/UDP).
direction	iPerf measurement direction (uplink/downlink).
filesize	iPerf transfer size/configuration parameter.
packet_error_rate	iPerf packet error rate (test descriptor, often UDP).
retransmissions	TCP retransmissions observed during the test.
parallel_streams	Number of parallel iPerf streams used.
tcp_max_cwnd_0	TCP maximum congestion window (low-level TCP metric).
tcp_mean_rtt_0	TCP mean RTT during the iPerf test (low-level TCP metric).
datarate	Measured throughput (Mbps).
network	Technology label (LTE/NR).
cell_index	Internal serving index.
physical_cellid	Serving PCI.
tracking_area_code	Serving TAC.
earfcn	Serving carrier frequency indicator.
rsrp	Serving RSRP recorded alongside iPerf measurement.

Column	Definition
rsrq	Serving RSRQ recorded alongside iPerf measurement.
rssi	Serving RSSI recorded alongside iPerf measurement.
sinr	Serving SINR recorded alongside iPerf measurement.
ta	Timing advance recorded alongside iPerf measurement.
cqi	CQI recorded alongside iPerf measurement.
primary_bandwidth	DL bandwidth configuration recorded alongside iPerf measurement.
cellbandwidths	Aggregated bandwidth recorded alongside iPerf measurement.
ul_bandwidth	UL bandwidth recorded alongside iPerf measurement.
mcc	MCC (operator metadata).
mnc	MNC (operator metadata).
ss_rsrp	SS-RSRP recorded alongside iPerf measurement.
ss_rsrq	SS-RSRQ recorded alongside iPerf measurement.
ss_sinr	SS-SINR recorded alongside iPerf measurement.
latitude	GPS latitude recorded alongside iPerf measurement.
longitude	GPS longitude recorded alongside iPerf measurement.
altitude	GPS altitude recorded alongside iPerf measurement.
location_accuracy	GPS accuracy recorded alongside iPerf measurement.
velocity	Speed recorded alongside iPerf measurement.
velocity_accuracy	Speed accuracy recorded alongside iPerf measurement.
bearing	Bearing recorded alongside iPerf measurement.
bearing_accuracy	Bearing accuracy recorded alongside iPerf measurement.
timestamp_day	Day extracted from timestamp (campaign context).
week_of_year	Week-of-year extracted from timestamp (campaign context).
day_of_week	Day-of-week extracted from timestamp (campaign context).
old_5_rsrp	Short-window historical/smoothed RSRP feature.
old_50_rsrp	Long-window historical/smoothed RSRP feature.
old_5_rsrq	Short-window historical/smoothed RSRQ feature.
old_50_rsrq	Long-window historical/smoothed RSRQ feature.
old_5_ss_rsrp	Short-window historical/smoothed SS-RSRP feature.
old_50_ss_rsrp	Long-window historical/smoothed SS-RSRP feature.
old_5_ss_rsrq	Short-window historical/smoothed SS-RSRQ feature.
old_50_ss_rsrq	Long-window historical/smoothed SS-RSRQ feature.
device	Device identifier/model (bias check; exclude from model inputs).

Static

Table 31: Static – `cell_data`: one-line feature dictionary

Column	Definition
id	Unique row identifier (static logs).
session_id	Measurement session identifier.
timestamp	Measurement timestamp.
network	Radio access technology label (LTE/NR).
cell_index	Internal log index for the serving cell.
physical_cellid	Serving cell identifier (PCI).
tracking_area_code	Tracking Area Code (TAC) for coarse region context.
earfcn	Carrier frequency/channel indicator.
rsrp	Serving-cell signal strength (RSRP, dBm).
rsrq	Serving-cell signal quality (RSRQ, dB).
rssi	Total received power (RSSI, dBm).
sinr	Signal-to-interference-plus-noise ratio (SINR, dB).
ta	Timing Advance (distance proxy / cell-edge indicator).
cqi	Channel Quality Indicator (UE feedback for achievable rate).
primary_bandwidth	Primary downlink bandwidth configuration.
cellbandwidths	Aggregated bandwidth/carrier configuration(s).
ul_bandwidth	Uplink bandwidth configuration.
lte_mcs	LTE modulation and coding scheme index (link adaptation).
lte_ri	LTE rank indicator (MIMO rank).
nr_mcs	NR modulation and coding scheme index (link adaptation).
nr_ri	NR rank indicator (MIMO rank).
tx_power	UE transmit power (uplink stress indicator).
mcc	Mobile Country Code (operator metadata).
mnc	Mobile Network Code (operator metadata).
ss_rsrp	NR synchronization signal strength (SS-RSRP).
ss_rsrq	NR synchronization signal quality (SS-RSRQ).
ss_sinr	NR synchronization signal SINR (SS-SINR).
device	Device identifier/model (bias check; exclude from model inputs).
devicename	Device model/name (metadata).
twog_possible	2G capability/availability flag.
threeg_possible	3G capability/availability flag.
fourg_possible	4G capability/availability flag.
fiveg_nsa_possible	5G NSA capability/availability flag.
fiveg_sa_possible	5G SA capability/availability flag.
username	Static site identifier (joins to `static_locations`).
session_start_timestamp	Session start timestamp (grouping metadata).

Column	Definition
MNO	Operator label (metadata).

Table 32: Static – `latency_data`: one-line feature dictionary

Column	Definition
refsig_id	RTT measurement/test identifier.
timestampstart	RTT test start timestamp.
packet_loss	Packet loss percentage during the ping window.
server_ip	Ping server IP address (configuration metadata).
mean_latency	Mean RTT during the ping window (ms).
min_latency	Minimum RTT during the ping window (ms).
max_latency	Maximum RTT during the ping window (ms).
mean_dev_latency	RTT deviation/jitter metric for the ping window.
no_pings	Number of pings attempted in the window.
device_id	Device identifier (metadata).
session_id	Measurement session identifier.
timestamp	Timestamp used for aligning QoS with radio measurements.
network	Technology label (LTE/NR).
cell_index	Internal serving index.
physical_cellid	Serving PCI.
tracking_area_code	Serving TAC.
earfcn	Serving carrier frequency indicator.
rsrp	Serving RSRP recorded alongside latency measurement.
rsrq	Serving RSRQ recorded alongside latency measurement.
rssi	Serving RSSI recorded alongside latency measurement.
sinr	Serving SINR recorded alongside latency measurement.
ta	Timing advance recorded alongside latency measurement.
cqi	CQI recorded alongside latency measurement.
primary_bandwidth	DL bandwidth configuration recorded alongside latency measurement.
cellbandwidths	Aggregated bandwidth recorded alongside latency measurement.
ul_bandwidth	UL bandwidth recorded alongside latency measurement.
lte_mcs	LTE MCS recorded alongside latency measurement.
lte_ri	LTE RI recorded alongside latency measurement.
nr_mcs	NR MCS recorded alongside latency measurement.
nr_ri	NR RI recorded alongside latency measurement.
tx_power	UE TX power recorded alongside latency measurement.
mcc	MCC (operator metadata).

Column	**Definition**
mnc	MNC (operator metadata).
ss_rsrp	SS-RSRP recorded alongside latency measurement.
ss_rsrq	SS-RSRQ recorded alongside latency measurement.
ss_sinr	SS-SINR recorded alongside latency measurement.
device	Device identifier/model (bias check; exclude from model inputs).
devicename	Device model/name (metadata).
twog_possible	2G capability/availability flag.
threeg_possible	3G capability/availability flag.
fourg_possible	4G capability/availability flag.
fiveg_nsa_possible	5G NSA capability/availability flag.
fiveg_sa_possible	5G SA capability/availability flag.
NBIOT_possible	NB-IoT capability/availability flag.
username	Static site identifier (joins to `static_locations`).
session_start_timestamp	Session start timestamp (grouping metadata).
MNO	Operator label (metadata).

Table 33: Static – `static_locations`: feature dictionary

height**Column**

Definition

username
Static site identifier used to join fixed coordinates to static measurements.
latitude
GPS latitude of the static measurement point.
longitude
GPS longitude of the static measurement point.
altitude
GPS altitude/elevation of the static measurement point.

General Conclusion

References

[1] A. H. Al-Bayatti et al. Mobility management for lte and beyond. *IEEE Communications Surveys Tutorials*, 2012.

[2] Cadence PCB Solutions. Urllc use cases in healthcare and beyond, 2023. Accessed: 2026-02-11.

[3] IEEE ComSoc Technology Blog. Ericsson on 5g use cases: remote surgery, augmented and virtual reality use cases, 2024. Accessed: 2026-02-11.

[4] Inside Telecom. 5g medical applications in healthcare in remote robotic surgery, 2024. Accessed: 2026-02-11.

[5] Netflix Help Center. Internet connection speed recommendations. Accessed: 2026-02-11.

[6] H. Soliman et al. Analysis of handover delay and packet loss in mobile ipv6. *IEEE Transactions on Vehicular Technology*, 2016.

[7] Y. Wang and B. Zong. Performance analysis of handover management schemes in high-speed railways. *IEEE Access*, 7:126892–126903, 2019.

Table 5: H-Bahn `cell_data`: simplified feature dictionary (serving cell)

Column	**Simple meaning**	**Why it matters for handover/QoS**
timestamp	Time of the measurement	Aligns radio samples with latency/throughput and supports "before/after HO" windows.
physical_cellid	ID of the serving cell (PCI)	Detects serving-cell changes (handover) and supports ping-pong analysis.
cell_index	Internal serving-cell index in the logs	Helps track serving-cell sequence within the log.
tracking_area_code	Tracking area ID (TAC)	Coarse mobility context (moving across areas can change network behavior).
network	Technology label (LTE/NR)	LTE and 5G NR can behave differently; can be modeled separately if needed.
earfcn	Carrier/frequency indicator	Frequency band affects coverage and explains different signal regimes.
rsrp	**Signal strength**	Low RSRP is typical at cell edges and increases handover risk and QoS degradation.
rsrq	**Signal quality** (incl. interference/load)	Explains poor experience even when RSRP is acceptable.
sinr	**Link cleanness** (signal vs interference)	Strong driver of throughput/latency stability; low SINR often precedes degradation.
rssi	Total received power	Helps interpret whether low SINR comes from interference or weak signal.
ta	Distance proxy (Timing Advance)	Highlights cell-edge conditions where handovers are less stable.
cqi	Channel quality indicator	Proxy for achievable data rate; supports QoS prediction.
tx_power	UE transmit power	High values can indicate challenging radio conditions (often uplink-related).
primary_bandwidth	Main downlink bandwidth configuration	Sets a capacity ceiling; helps explain throughput differences.
ul_bandwidth	Uplink bandwidth configuration	Provides uplink capacity context; can indirectly affect perceived performance.
cellbandwidths	Bandwidth configuration(s) across carriers	Indicates multi-carrier / aggregated bandwidth context and capacity differences.

Table 6: H-Bahn `neighboring_data`: simplified feature dictionary (candidate neighbor cells)

Column	**Simple meaning**	**Why it matters for handover/QoS**
physical_cellid_neighboring	ID of a neighbor cell (candidate target)	Core input for DSO2: defines the candidate set of target cells.
earfcn_neighboring	Neighbor frequency indicator	Helps compare candidates operating on different bands.
tracking_area_code_neighboring	Neighbor TAC	Adds regional context for the neighbor cell.
network_neighboring	Neighbor technology label	Helps compare candidates across LTE/NR contexts.
rsrp_neighboring	Neighbor signal strength	Basic indicator of whether the neighbor is a plausible target.
rsrq_neighboring	Neighbor signal quality	Avoids targets with strong power but poor quality (interference/load).
sinr_neighboring	Neighbor SINR	Proxy for expected QoS after switching to that neighbor.
rssi_neighboring	Neighbor total received power	Helps interpret interference conditions for the neighbor.
latitude, longitude, altitude	GPS position at measurement time	Maps where certain neighbors become good/bad candidates along the route.
velocity, bearing	Mobility context at that time	Candidate quality can depend on speed and direction (trajectory segment).
location_accuracy, velocity_accuracy, bearing_accuracy	Sensor accuracy estimates	Used to filter unreliable mobility/location readings.
mcc_neighboring, mnc_neighboring, MNO_neighboring	Operator identifiers (metadata)	Mainly metadata/sanity checks; usually not used as predictive drivers.
device	Device identifier/model	Bias control; can be excluded to avoid device-specific leakage.

Table 7: Feature inclusion policy used in this project

Group	Examples	Inclusion rationale
Core (always used)	`rsrp, rsrq, sinr, ta, cqi, velocity`	Strong domain link to coverage/interference/mobility and generally available; supports robust modeling across scenarios.
Optional (used if available)	`ss_rsrp, ss_rsrq, ss_sinr, tx_power, lte_mcs, nr_mcs, lte_ri, nr_ri`	Informative but more missing or scenario-dependent; included as extensions and handled with imputation/missingness indicators.
Neighbor-only (DSO2)	`rsrp_neighboring, rsrq_neighboring, sinr_neighboring, physical_cellid_neighboring`	Required to rank candidate target cells; cannot be replaced by serving-only data.
QoS outcomes (targets)	`mean_latency, packet_loss, datarate`	Direct user-perceived outcomes used as regression targets and for QoS-break proxies.
Excluded (metadata/leakage)	`device, mcc, mnc, MNO, server_ip, passive_id, username`	Either identifiers, operator metadata, or experiment configuration; excluded to prevent leakage and improve generalization.

Table 8: Selected key variables and short justification (used for EDA and later modeling)

Selected feature	Reason (simple)	Supports
rsrp, rsrq, sinr	Core radio indicators: strength + quality + interference; directly relate to HO stability and QoS.	DSO1, DSO4
ta	Distance proxy; highlights cell-edge conditions where HO is risky.	DSO1
cqi	Channel quality feedback; proxy for achievable throughput.	DSO4
velocity, bearing	Mobility regime; affects HO timing and route-dependent behavior.	DSO1, DSO3
latitude, longitude	Spatial context to locate recurring problem zones.	DSO3
physical_cellid	Identifies serving cell to extract HO transitions and ping-pong events.	DSO2 (prep), DSO3
rsrp_neighboring, rsrq_neighboring, sinr_neighboring	Neighbor quality to rank candidate target cells.	DSO2
physical_cellid $_{n}eighboring$	Identifies each neighbor candidate for ranking.	DSO2
mean_latency, packet_loss	QoS measures used for latency SLA compliance and QoS-break proxies (targets/labels).	DSO4, DSO1 (proxy)
datarate	iPerf throughput proxy for user-perceived throughput.	DSO4

Table 9: Top missingness and uniqueness for candidate features, H-Bahn (`latency_data`)

Feature	Missing (%)	n_unique
network	100.00	0
cellbandwidths	100.00	0
rssi	43.97	29
rsrq	5.57	14
ss_sinr	2.02	41
ss_rsrq	2.01	13
ss_rsrp	2.01	59
tx_power	1.37	43
ul_bandwidth	0.42	2
primary_bandwidth	0.42	2

Table 10: Top missingness and uniqueness for candidate features, Mobile (`latency_data`)

Feature	Missing (%)	n_unique
network	100.00	0
ul_bandwidth	100.00	0
ss_rsrp	44.43	54
ss_sinr	44.43	26
ss_rsrq	44.43	14
cellbandwidths	0.45	12
rsrq	0.13	17
velocity	0.07	2819
sinr	0.00	38
rsrp	0.00	58

Table 11: Example of highly correlated feature pairs removed.

Scenario	Feature 1	Feature 2	$\lvert\lvert$
H-Bahn	rsrp	ss_rsrp	0.86
H-Bahn	rsrq	sinr	0.85
Mobile	rsrp	rssi	0.94

Table 12: Feature-group vs. DSO traceability (initial hypothesis)

Feature Group	**DSO1**	**DSO2**	**DSO3**	**DSO4**
Serving-cell radio (RSRP/RSRQ/SINR, SS-*)	Yes	Support	Yes	Support
Mobility/context (velocity, GPS, time)	Yes	Support	Yes	Support
Neighbor features (neighbor RSRP/RSRQ/SINR, deltas, PCI)	Support	Yes	Support	Support
QoS measurements (RTT stats, packet loss, iPerf datarate)	Support (proxy)	Support	Support	Yes

Table 13: Key variables and their descriptions (used throughout Phase 2)

Variable	**Unit**	**Description**
rsrp	dBm	Serving-cell signal strength.
rsrq	dB	Serving-cell signal quality (includes interference/load effects).
sinr	dB	Serving-cell interference indicator (signal vs. interference+noise).
ss_rsrp	dBm	5G NR synchronization signal strength.
ss_sinr	dB	5G NR synchronization signal SINR.
velocity	m/s	User speed (from GPS).
mean_latency	ms	Average RTT (latency) for the measurement window.
packet_loss	%	Packet loss during the measurement window.
datarate	Mbps	Throughput measured via iPerf.
rsrp_neighboring	dBm	Neighbor-cell signal strength.
physical_cellid	–	Serving cell identifier (PCI).
physical_cellid_neighboring		Neighbor cell identifier (PCI).

Table 14: Key variables and their descriptions (used throughout Phase 2)

Variable	**Unit**	**Description**
rsrp	dBm	Serving-cell signal strength.
rsrq	dB	Serving-cell signal quality (includes interference/load effects).
sinr	dB	Serving-cell interference indicator (signal vs. interference+noise).
ss_rsrp	dBm	5G NR synchronization signal strength.
ss_rsrq	dB	5G NR synchronization signal quality.
ss_sinr	dB	5G NR synchronization signal SINR.
tx_power	dBm	Device transmit power; indicator of uplink radio environment.
earfcn	–	LTE absolute frequency channel number; differentiates band-specific behavior.
rssi	dBm	Received signal strength indication; broad total input power measure.
lte_mcs	–	LTE modulation/coding scheme; reflects link adaptation and throughput potential.
nr_mcs	–	NR modulation/coding scheme.
velocity	m/s	User speed (from GPS).
velocity_accuracy	m/s	Precision of velocity estimate; used for quality control.
mean_latency	ms	Average RTT (latency) for the measurement window.
packet_loss	%	Packet loss during the measurement window.
datarate	Mbps	Throughput measured via iPerf.

Table 15: Missing Values Summary by Scenario

Scenario	**File**	**Missing Values**	**Missing (%)**
H-Bahn	cell_data	312,450	8.9
	neighboring_data	45,230	2.3
	latency_data	18,540	0.4
	iperf_data	52,180	1.6
Mobile	cell_data	285,670	8.7
	neighboring_data	38,920	1.7
	latency_data	22,450	0.5
	iperf_data	41,280	0.8
Static	cell_data	456,230	12.3
	latency_data	28,670	0.6

Table 20: DSO4 — Model Hyperparameters

Parameter	**Random Forest**	**HistGradBoost**	**Gradient Boosting**
n_estimators / max_iter	300	500	300
max_depth	None	8	6
learning_rate	—	0.05	0.05
max_features	0.6	—	—
min_samples_leaf	3	20	—
subsample	—	—	0.8
l2_regularization	—	0.1	—
n_jobs	-1	—	—
random_state	42	42	42

Table 21: DSO4 — Post-Handover Throughput Prediction: Regression Metrics (Test Set)

Model	**MAE (Mbps)**	**RMSE (Mbps)**	$\mathbf{R^2}$
Random Forest (tuned)	4.157	8.685	0.9927
HistGradientBoosting	5.147	8.197	0.9935

Table 22: DSO4 — Post-Handover Throughput Prediction: Tolerance Accuracy (Test Set)

Model	**±10% Acc**	**±20% Acc**	**±10 Mbps**	**±20 Mbps**
Random Forest (tuned)	75.05%	85.17%	90.38%	96.63%
HistGradientBoosting	64.86%	80.87%	87.78%	97.26%

dso4_model_comparison.png

Figure 11: DSO4 — Model comparison across MAE (Mbps), RMSE (Mbps), and R^2 metrics.

dso4_feature_importance.png

Figure 12: DSO4 — Top 15 feature importances from the Random Forest model (mean decrease in impurity).